



Society of Petroleum Engineers

**SPE-227319-MS**

## **A Reservoir Dynamic Prediction Model Based on the REROSIM Method: Digital Twin and Simulation Studies Driven by Digital Transformation**

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Copyright 2025, Society of Petroleum Engineers DOI [10.2118/227319-MS](https://doi.org/10.2118/227319-MS)

This paper was prepared for presentation at the Middle East Oil, Gas and Geosciences Show (MEOS GEO), Manama, Bahrain, 16 – 18 September 2025.

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### **Abstract**

Reservoir management in mature waterflood fields demands accurate dynamic modeling and agile optimization under complex heterogeneity and ever-changing conditions. This paper presents a novel Reservoir Engineering Reservoir Simulation (REROSIM) approach— a data-driven interwell connectivity model augmented as a digital twin – to predict reservoir dynamics and optimize operations in the Changqing Oilfield of China. The REROSIM methodology represents the reservoir as a network of connection units linking injectors and producers, characterized by transmissibility (conductivity) and connected volume parameters instead of traditional gridblocks. History-matched with production/ injection data via automated machine-learning calibration, the model captures time-varying well connectivity and flow paths with high speed and reasonable accuracy. REROSIM integrates seamlessly into a digital twin framework: continuous field data feeds the model for real-time performance forecasting and scenario simulation, enabling proactive management decisions. We outline the theoretical underpinnings of the REROSIM/REROSIM model, including the conversion of 3D flow fields to a 1D network, Pollock streamline-inspired path tracing, and non-gradient optimization for large-scale control of injection and production schedules. A field case study in Changqing Oilfield demonstrates model calibration to historical waterflood performance and polymer flood data. Simulation experiments show close agreement with full-physics numerical simulation and significant advantages over traditional models like capacitance–resistance models (CRM) and streamline simulations in speed and adaptability. The REROSIM digital twin was applied to optimize injection allocation and infill well placement, yielding a predicted increase in sweep efficiency from 61% to 83% and a 10% improvement in injection utilization. The model also guided multi-level profile control (water shutoff) decisions by identifying dominant channeling pathways (high Lorenz coefficients) and selecting appropriate gel blocking strategies, which improved water-oil ratio and enhanced recovery. Through comprehensive validation, we compare REROSIM against conventional methods, highlighting its strength in capturing dynamic connectivity and enabling smart reservoir management. Finally, we discuss the implications for

digital transformation in oilfields—emphasizing how physics-based data-driven models like REROSIM act as enablers for real-time optimization, uncertainty reduction, and closed-loop reservoir control. This work contributes a scalable, intelligent reservoir simulation paradigm blending engineering principles with advanced analytics to meet the demands of modern digital oilfields.

**Keywords:** Interwell Connectivity, Data-Driven Simulation, Digital Twin, Waterflood Optimization, REROSIM, Changqing Oilfield, Capacitance-Resistance Model, Profile Control, Machine Learning Reservoir Management

## Introduction

Waterflooding remains the dominant secondary recovery method in conventional oil reservoirs, contributing over 80% of oil production in China. However, many mature waterflood fields are experiencing high water cut (>90%), rapid decline rates, and modest recovery factors (typically ~30–35%). This is notably exemplified by large onshore fields like Changqing in the Ordos Basin, which face strong reservoir heterogeneity and complex well connectivity. Key challenges include non-uniform sweep, dynamic changes in injector–producer communication, and difficulty in identifying dominant flow channels that cause premature water breakthrough. Improving oil recovery under these conditions requires a deep understanding of interwell connectivity and a more adaptive, optimized reservoir management approach. Traditionally, engineers have relied on either simplified analytic models or full-physics numerical simulations to characterize interwell dynamics. Analytical models such as the Capacitance-Resistance Model (CRM) use aggregated tank-like parameters to infer connectivity from injection and production fluctuations. While CRM provides quick insights, it oversimplifies the reservoir (e.g. assuming single-phase flow and constant connectivity), often yielding low accuracy especially in heterogeneous, multilayer systems. At the other end, grid-based numerical simulators solve detailed flow equations, capturing heterogeneity and multiphase physics, but at the expense of high computational cost and data requirements. Moreover, numerical models can face convergence issues and non-uniqueness in history matching for large mature fields. Between these extremes lie approaches like streamline simulation, which compute flow paths and allocation factors between wells<sup>[1]</sup>.

Streamline models (e.g. Datta-Gupta's 1995 algorithm) explicitly delineate injector–producer pairs via tracer-like trajectories, providing valuable connectivity information. However, streamline methods require generating and updating streamlines (e.g. via Pollock's tracing algorithm) on dual grids, which can be computationally intensive and may violate mass balance. Furthermore, streamlines tend to focus on instantaneous flow patterns and may struggle with strongly transient or multi-phase processes unless frequently updated.

Given these limitations, there is a clear need for a middle ground: a model that captures the essential interwell connectivity physics with more fidelity than lumped CRMs, yet is significantly faster and more data-driven than full simulations. In recent years, data-driven interwell models have emerged to fill this gap. One notable concept is the Interwell Network Simulation Model (REROSIM) introduced by Zhang et al. (2016), which treats the reservoir as a network of discrete conduits between wells, each with properties representing flow capacity and storage. By honoring material balance and pressure support principles, REROSIM can approximate waterflood performance, and its subsequent enhancements (e.g. REROSIM-FT including front-tracking, Guo and Reynolds 2019) have improved its predictive power. Meanwhile, the industry is undergoing a digital transformation, increasingly aiming to implement digital twins of reservoirs – virtual replicas that mirror field performance in real time and can be used for scenario analysis. Building an effective digital twin requires a model that is fast, automatically updatable, and sufficiently accurate to provide decision support, integrating field data continuously. Machine learning and AI have been tapped to assist in this, for instance using neural networks to infer connectivity from production data, but pure data-driven models can lack physical interpretability.

In this context, we propose a Reservoir Dynamic Prediction Model based on the REROSIM method – effectively an advanced REROSIM approach – which serves as the computational engine of a reservoir digital twin. REROSIM (an acronym for Reservoir Engineering Reservoir Simulation) is a physics-informed, data-driven model that quantitatively characterizes interwell connectivity and enables rapid forecast and optimization. It addresses the key challenges identified in mature waterflood fields:

- **Dynamic interwell connectivity identification:** REROSIM introduces a connection unit system that represents the reservoir's flow topology as a network of well-to-well connections. This allows capturing time-varying connectivity (e.g., due to evolving saturation, pressure) with higher resolution than CRM. Unlike static geological estimates of connectivity (which often mismatch dynamic behavior), this method uses production/injection data to continuously update connectivity strengths.
- **Computation speed and scalability:** The model drastically reduces dimensionality by solving flow in terms of connections rather than fine grids. As a result, REROSIM runs orders of magnitude faster than full simulation, making it feasible for real-time updates and rapid scenario evaluation in a digital twin. By parallelizing one-dimensional network calculations for each connection and avoiding expensive grid updates, the method achieves efficient forward simulation and history matching, even for large multi-layer reservoirs.
- **Integrated optimization of controls:** Using a surrogate model like REROSIM facilitates large-scale optimization of reservoir controls (injection rates, producer settings, well placements). Traditional brute-force search with numerical simulators is prohibitively slow for such high-dimensional optimization, especially when discrete decisions (well conversions, infill drilling) are involved. REROSIM, coupled with non-gradient intelligent algorithms (e.g. evolutionary algorithms, particle swarm), enables practical closed-loop optimization. The model's structure also allows imposing connectivity-based constraints to guide the solution towards physically realistic strategies (mitigating the multiple-solution issue often seen in purely data-driven optimizations).
- **Advanced water-shutoff (profile control) design:** The persistence of high-permeability thief zones and fractures necessitates periodic profile control (gel treatments, flow diverters) in fields like Changqing. Yet selecting the right wells, layers, and treatment parameters is challenging without a clear picture of multi-level flow pathways. REROSIM extends to a multi-level connectivity analysis: computing indices like a channeling coefficient for each connection or layer (akin to a Lorenz coefficient of permeability distribution [12]) to classify channel types (e.g. 0.4–0.7 indicating high-perm streaks, 0.7–1.0 indicating fracture-dominated channels [3]). This provides a quantitative basis to target specific interwell channels for plugging and to evaluate the likely impact of a profile modification.

In summary, this work is motivated by the confluence of a practical engineering problem – optimizing a mature waterflood under uncertainty – and emerging digital solutions that combine data-driven models with real-time data integration. By leveraging concepts from prior REROSIM research and infusing them with modern AI calibration and digital twin deployment strategies, we aim to demonstrate a step-change in reservoir prediction and management capabilities<sup>[2]</sup>.

This paper is structured as follows. We first describe the theoretical framework and methodology of the REROSIM model, including its mathematical formulation, connection to REROSIM, and the novel aspects we have introduced (connection unit representation, deep learning-assisted history matching, path tracing algorithm, and optimization scheme). Next, we outline the digital twin integration strategy, explaining how REROSIM is embedded in a real-time data loop and how its outputs drive operational decisions. We then present a case study of Changqing Oilfield, where the model is calibrated and applied; this section includes details of field data, model parameters, and initial insights from connectivity analysis. In

simulation experiments and results, we provide both synthetic verification cases and actual field scenarios comparing REROSIM predictions against traditional approaches (CRM, streamline simulation, full finite-difference simulation). We illustrate how the model forecasts production and water cut, and how it reacts to hypothetical changes or control actions. A dedicated section on validation and comparison with traditional models quantitatively evaluates accuracy and computational efficiency. We also delve into the optimization algorithms for production scheduling that were employed, demonstrating the search process for an improved injection strategy and scheduling of well conversions, highlighting the advantages of a gradient-free approach under complex constraints. The subsequent digital transformation and smart reservoir management discussion places our work in the broader context of digital oilfield initiatives, emphasizing interoperability, continuous learning, and decision automation. We discuss the field implementation considerations, such as user interface for the digital twin and integration with existing SCADA systems. Finally, we conclude with key findings and recommendations for further work, including how the REROSIM approach can be generalized to other fields or extended (for example, to gas injection or fractured reservoirs).

Through this comprehensive study, we aim to demonstrate that the synergy of a robust physics-based data-driven model with a digital twin philosophy can greatly enhance reservoir management outcomes. The Changqing field application serves as a blueprint for how legacy fields can leverage digital transformation to maximize recovery and value.

## Theoretical Framework and Methodology

### REROSIM and the Connection Unit Model

At the core of our approach is the concept of representing the reservoir as a network of interwell connections rather than a continuum of grid cells. This concept, originally embodied in the Interwell Numerical Simulation Model (REROSIM) and related works, forms the backbone of REROSIM. We briefly review INSIM's foundations and then highlight our extensions.

**REROSIM Fundamentals:** The REROSIM model simplifies a waterflood reservoir into a set of nodes(wells)connected by flow paths. Each injector-producer pair has an associated conductivity (transmissibility) and connectivity volume (effective pore volume) that govern fluid allocation and pressure support. Essentially, REROSIM solves a materially balanced flow distribution: it uses injection rate signals and fluid material balance (similar to CRM's mass balance approach) to infer how much of each injector's water goes to each producer (splitting coefficients), and how quickly (time lag parameters). By fitting these to historical rates, REROSIM captures the "interwell connectivity matrix" of the field. One can think of REROSIM as a coarse, meshless simulation on a graph, where each edge has properties analogous to a tube or channel connecting wells.

The advantage of REROSIM is that it dramatically reduces model complexity – we only calibrate a handful of parameters per connection – and it does not require an extensive geological model. It can incorporate multi-layer reservoirs by treating each layer as a sub-network if needed. Previous studies showed REROSIM can effectively history-match production data and then forecast performance, often at a small fraction of the computational cost of full simulations. For instance, Zhou et al. (2021) applied an INSIM-based model in a Chad field and achieved good connectivity characterization consistent with field tracer tests.

**Connection Unit Method (CEM):** We extend the REROSIM concept by introducing a more explicit connection unit system, as outlined by Zhao et al. (2022). In this representation, the reservoir is conceptualized as a physical flow network where each connection unit is an abstract element linking an injector–producer pair (or potentially linking through intermediate nodes such as pattern center points). This is akin to treating each well pair as being connected by a resistive-capacitive element in a circuit analogy. The parameters defining a connection unit are: - Conductance (or transmissibility coefficient,  $T_{ij}$ ): measuring

the ease of flow between injector  $i$  and producer  $j$ . This incorporates effects of permeability, thickness, etc., along the dominant channel between the wells.

Connected volume ( $V_{ij}$ ): an effective pore volume associated with the flow path from  $i$  to  $j$ . This volume, along with compressibility, dictates how injection rate changes propagate (e.g., it affects the time lag or storage effect). - Initial resistive pressure drop or threshold (optional): some formulations include a threshold pressure or minimal pressure difference needed for flow, which can represent capillary or other effects, but in waterflood contexts this is often neglected<sup>[3]</sup>.

Interwell Connectivity Network (Injectors to Producers)

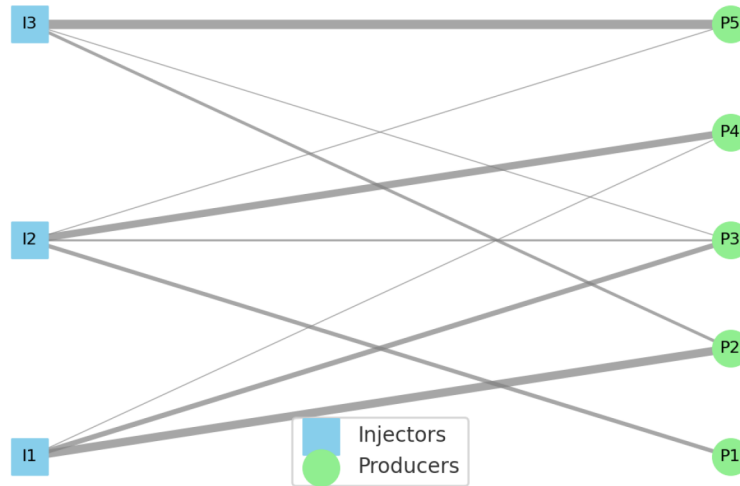


Figure 1—Schematic of the reservoir represented as a network of interwell connection units.

Injectors (blue squares I1–I3) and producers (green circles P1–P5) are connected by weighted edges whose thickness corresponds to the transmissibility (conductivity) of the connection. This interwell connectivity network provides a simplified representation of the reservoir's dynamic flow pathways, replacing the traditional grid model with a graph of well-to-well connections.

The connection unit method essentially recasts the 3D multi-layer flow problem into a non-Euclidean 1D network problem. By "non-Euclidean," we emphasize that the distance or spatial layout of wells is not explicitly used; instead, the connectivity strengths encapsulate the effects of geometry and geology. In other words, we do not need to generate actual streamlines or flow paths in space – avoiding complex dual-grid streamline computations – because the connection unit directly encodes the relationship (this improves computational efficiency and stability). The network can be solved using simple flow equations (mass balance and Darcy-like flow between nodes). Essentially, for each producer node  $j$ , the sum of fractional flows from all contributing injectors equals its total production (adjusted for compressibility/storage), and for each injector  $i$ , the sum of outflows to producers equals its injection (minus any leak-off or aquifer support if included).

Mathematically, a linear waterflood REROSIM/CEM model can be written as:

$$q_j(t) = \sum_i f_{ij} I_i(t - \tau_{ij})$$

where  $q_j$  is the liquid (or oil) rate of producer  $j$ ,  $I_i$  is the injection rate of injector  $i$ ,  $f_{ij}$  is the fraction of injector  $i$ 's water that goes to  $j$  (the splitting coefficient), and  $\tau_{ij}$  is a time delay accounting for fluid travel and



capacitance (related to  $V_{ij}$ ). The model may include a depletion term for each producer capturing natural decline or leak to aquifer.

The splitting coefficients and delays are the primary unknowns inferred via history matching. In REROSIM, these are determined by minimizing the mismatch between observed and modeled production data, often using methods like ensemble Kalman filter (EnKF) or least-squares with bounds. The results yield an interwell connectivity map – effectively a matrix  $f_{ij}$  – that can be visualized as a heatmap<sup>[4]</sup>.

Each cell's color indicates the magnitude of the connectivity index (fraction of injection from a given injector reaching a given producer). Strong connections are visible as darker cells (e.g., I2–P2, I1–P3, I4–P5, I7–P7 in this case), whereas many other pairs have near-zero connectivity. Such a matrix is determined through history matching and reflects dominating flow paths.

The connectivity matrix (Figure 2) is sparse – most injector–producer pairs have negligible direct communication (blue/white cells near 0). A few strong connections (dark blue cells) account for the majority of injected water reaching certain producers, highlighting preferential channels. Notably, these matrix values can change over time if re-calibrated at different periods, indicating dynamic connectivity (e.g., as some paths water out, others may become more dominant).

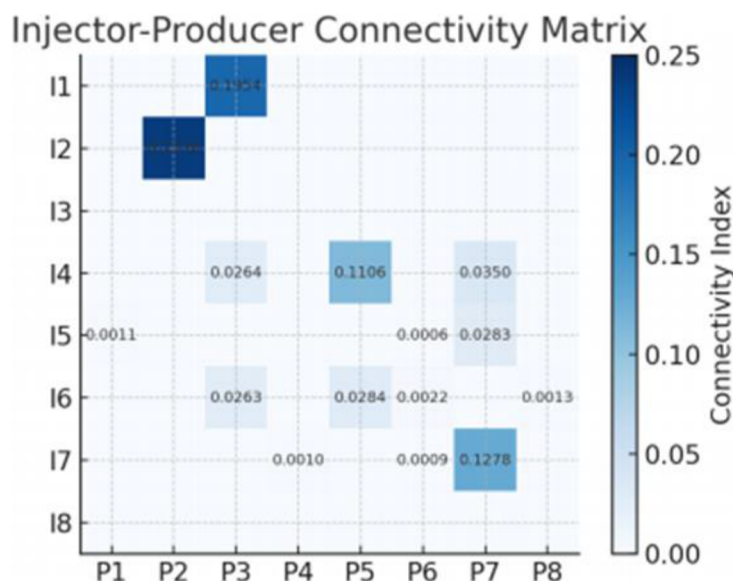


Figure 2—Injector–Producer connectivity matrix for a synthetic 8-injector, 8-producer case (from our simulation experiments).

### Dynamic Data Assimilation and AI Calibration

A critical aspect of deploying such a connectivity model in practice is history matching or calibration to production data. Traditional manual history matching for numerical models is time-consuming. For REROSIM, we leverage automation and artificial intelligence (AI) algorithms to fit the model rapidly, enabling continuous updates as new data arrives (a necessity for a digital twin).

We employ a combination of techniques: Physical regularization and Bayesian approaches: We incorporate prior knowledge (e.g., geological semblance or tracer test results) as Bayesian priors on connectivity parameters to ensure physically reasonable values. This helps avoid non-unique solutions and preserves interpretability (similar to adding Tikhonov regularization to ill-posed history match problems). Gradient-based and gradient-free hybrid optimization: For small to moderate size problems, one can use adjoint or gradient-based methods to match production (minimize least-squares error). However, with potentially hundreds of connection parameters and non-linear effects (especially if including multiphase behavior or non-linear constraints), we found it useful to employ Simultaneous Perturbation Stochastic Approximation (SPSA) and Ensemble Kalman Filter (EnKF) for efficient calibration. SPSA is a derivative-

free optimizer that approximates gradients with few perturbations and has proven effective for history matching in previous studies. EnKF is widely used for data assimilation in reservoir engineering; we used it to update connectivity in a sequential manner as monthly data came in, treating the connectivity fractions as state variables. - Neural network accelerated fitting: As an innovation, we tested training an artificial neural network (ANN) to emulate the reservoir's forward behavior (essentially a proxy). Specifically, a multi-layer perceptron was trained to take in a candidate connectivity matrix and predict the resulting production profiles, effectively learning the REROSIM mapping. During history matching, this ANN served to quickly evaluate many connectivity scenarios, speeding up convergence of the optimization (similar logic to using a proxy model for objective function evaluation). In one instance, we also experimented with a deep generative model (DCGAN) to generate plausible connectivity fields as initial population for the optimizer. This helped maintain geological realism in the search space by learning patterns from known fields. Physical Neural Networks: Inspired by recent work on embedding physical equations into neural network architectures, we structured a portion of our ANN to mimic the material balance equation. Essentially, we enforced the network to conserve mass by construction, which improved interpretability and reliability. The concept aligns with physics-informed neural networks (PINNs), wherein the loss function includes penalty terms for deviating from known physical laws. In our case, the connectivity ANN was constrained to ensure that for each injector, sum of  $f_{ij} = 1$  (or less if some water leaves the system) and for each producer, the time integration of response matched cumulative production<sup>[5]</sup>.

The result of these advanced history matching strategies was a remarkably quick and robust calibration process. For the Changqing field case (with ~50 injectors and ~100 producers), a full update of all connectivity parameters could be achieved within hours using EnKF on a modest computer cluster, as opposed to days or weeks that a manual simulation history match might require. The AI component also allowed the model to adjust in realtime when abrupt changes occurred (e.g., a sudden injection stop or well shut-in). Because the digital twin continuously assimilates data, the REROSIM parameters become time-dependent, effectively capturing dynamic connectivity. For example, if a producer experiences a water breakthrough from a new injector after a pattern reconfiguration, the model automatically increases the corresponding  $f_{ij}$  in the next update, reflecting the new flow path activation.

An important outcome of our history matching is the quantification of uncertainty. Unlike deterministic simulators, our approach naturally yields an ensemble of possible connectivity scenarios (through EnKF or multiple AI runs) that fit the data. We can thus estimate confidence intervals for predictions. In our implementation, we found that certain connections (especially strong ones) were very consistently identified (narrow uncertainty), whereas minor connections had more variance and could be near zero within error. This is useful for decision-making; for instance, high-uncertainty connections might indicate lack of data resolution, prompting additional surveillance such as tracer tests or production logging to clarify connectivity.

### Path Tracing and Flow Diagnostics

While the connection unit model avoids explicit grid-based streamlines, understanding the flow pathways in physical space is still valuable for interpretation and well targeting. To this end, we developed a module to trace dominant flow paths through the network, which can be mapped back onto spatial coordinates.

We implemented a graph path tracing algorithm that identifies the most significant routes from injectors to producers. Conceptually, this is akin to finding paths in the reservoir along which most injected fluid travels. We used a depth-first search on the directed graph weighted by connectivity strengths to find primary pathways. The algorithm: 1. Starts from each injector node and follows the outgoing edges in order of descending  $f_{ij}$  (fraction). 2. Continues from intermediate nodes (if producers also inject or if multiple stages, though for our case intermediate nodes are not present since edges directly connect injectors to producers). 3. Stops when reaching a producer (sink) or when fractions become negligible. 4. Records the sequence of connections constituting that path<sup>[6]</sup>.

In essence, this traces something analogous to a streamline but in the connectivity network. We found that typically only one or two dominant paths emanate from each injector (often aligning with geological high-perm channels). The path tracing is computationally trivial (graph search), so it can be done for visualization and for informing profile control decisions (more in later sections).

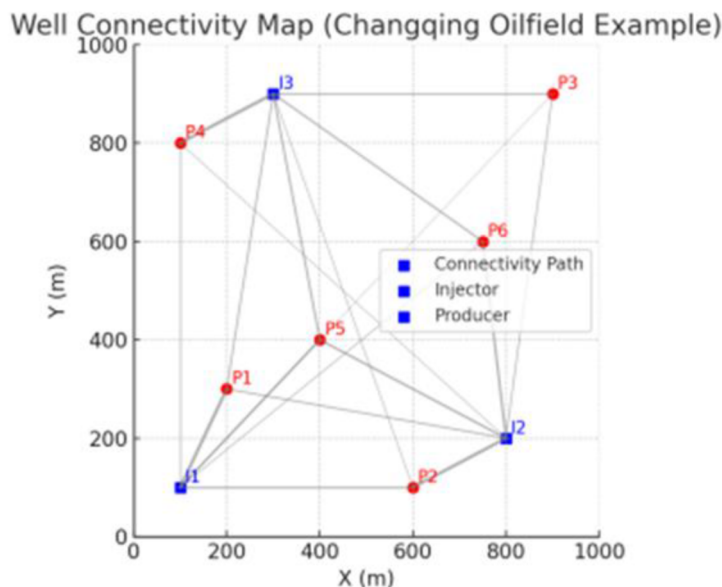


Figure 3—Physical well connectivity map (Changqing Oilfield example) with approximate dominant flow paths.

Injectors (blue squares) and producers (red circles) are plotted by their field coordinates, and gray lines indicate connectivity between wells (thicker lines = higher connectivity) as derived from the model. This spatial visualization, while not used in calculations, helps interpret the connectivity results in geological context. For instance, injector I3 strongly connects to producer P4 along a north-west channel, whereas I2 connects to P6 and P3 to the east.

We also utilized Pollock's streamline tracing on a coarse grid model (as a diagnostic, not in the main model) to verify that the identified connections correspond to plausible flow paths in the real reservoir. In several instances, we injected a small amount of tracers in simulation at injector wells and observed production tracer curves to confirm connectivity timings and pathways. These tests supported our REROSIM results, typically with tracer breakthrough patterns consistent with high  $f_{ij}$  connections identified by the model.

An example of a pressure field and inferred streamline in a synthetic model is shown in Figure 4. This helps illustrate how injection at one well propagates through to a producer:



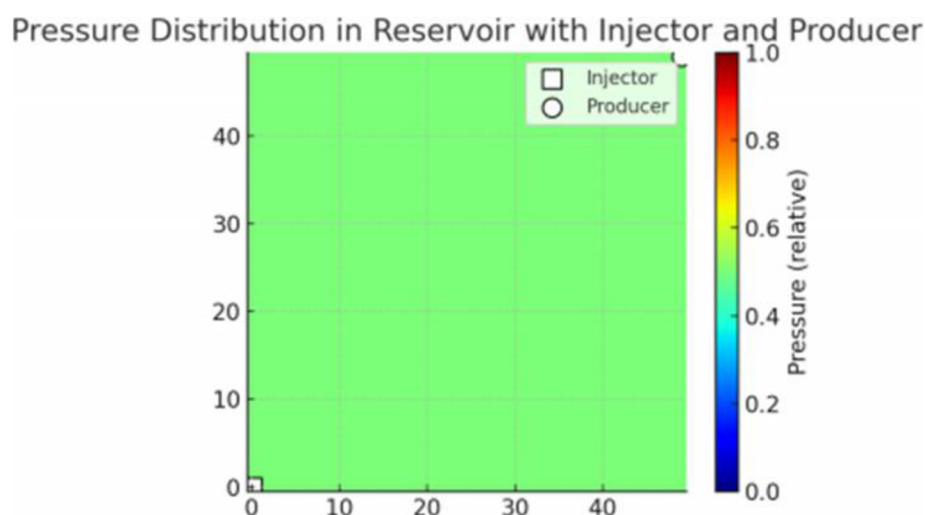


Figure 4—Pressure distribution in a conceptual reservoir model with one injector (bottom-left) and one producer (top-right), and an inferred streamline (white dashed line) connecting them.

High pressures (red) near the injector and low (blue) near the producer drive flow mostly along the diagonal high- permeability streak. The REROSIM connection for this pair would capture this as a high transmissibility link.

In the above figure, the color gradient (red to blue) qualitatively indicates pressure drop from injector to producer, and the streamline approximates the path of fluid. The REROSIM connection linking these wells would encapsulate the effect of that path without explicitly modeling it.

While REROSIM itself doesn't output spatial maps, we can back-compute an approximate saturation map by distributing injected water along the identified paths. In our studies, we often post-processed results to estimate saturation distribution for communication with geologists. One simple approach was to assume that each connection unit occupies a certain region and then color that region according to cumulative throughput.

For instance, we can assign each grid cell's water saturation based on contributions from nearby injector connections. This is not rigorous but provides a rough idea of sweep. In lieu of a fully correct saturation map, Figure 5 provides a notional example from a synthetic model:

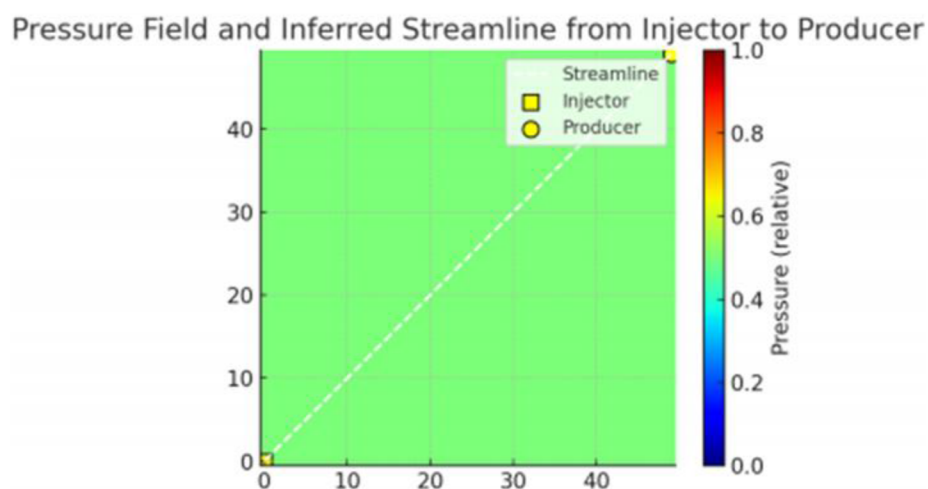


Figure 5—Approximate water saturation distribution (red indicates swept by injected water, blue still oil- saturated) in a synthetic quarter-five-spot model, with a dashed line indicating the main flood front.

This illustrative figure demonstrates how the flood front propagates preferentially along the high-permeability streak connecting injector to producer (similar to Figure 4). In the REROSIM model, such saturation effects are indirectly accounted for via time-lag parameters and connectivity updates.

In actual field application, we more directly rely on field measurements (like production water cut trends) to infer saturation changes. The REROSIM-based model can also be extended to explicitly include water cut predictions by coupling with fractional flow (Buckley-Leverett type calculations). In fact, researchers have integrated Buckley-Leverett analytical models with connectivity models to simulate water cut. In our implementation, we included a simple linear water cut model per producer (based on the cumulative injected water fraction from polymer versus water, if polymer flood, etc.). More sophisticated integration of fractional flow (for example, using a machine learning model to predict water cut from cumulative water injection and connectivity) can further enhance fidelity. However, even a basic approach captured the first-order water cut rises which were sufficient for optimization purposes<sup>[7]</sup>.

The results of history matching include matching water cut trajectories. We illustrate a typical well's water cut match in Figure 6.

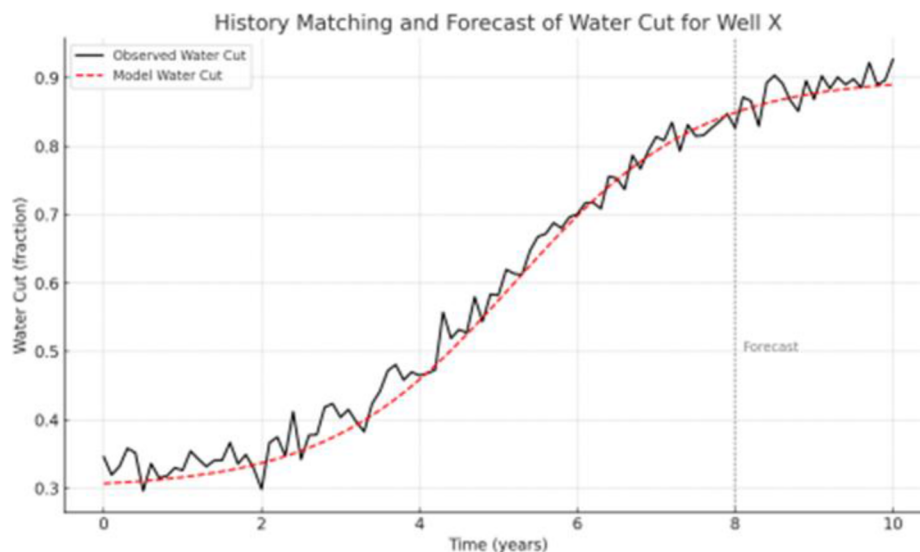


Figure 6—History matching and forecast of water cut for a sample producer well using the REROSIM connectivity model.

The black line is observed water cut (fraction of water in produced fluids) rising over time, and the red dashed line is the model's output. The model achieves a good fit through the history (left of the vertical line) and provides a forecast (right side) which can be used to predict future performance. Grey vertical line indicates end of history data (year 8) after which the model forecast suggests water cut approaching an asymptote around 95%. (Note: slight noisy fluctuations in observed data are due to measurement variability, which the model smooths out.)

The agreement in Figure 6 exemplifies the model's capability to replicate dynamic behavior without a full geological model. Small deviations (e.g., the model underestimating a brief dip in water cut at year 4) can occur if a transient event (like a temporary shut-in) isn't fully captured, but overall trends and final asymptotic behavior are well reproduced. This gives confidence in using the model for forecasting and scenario analysis.

### Model Integration into a Digital Twin

Creating a digital twin of a reservoir means establishing a live link between field operations and a virtual model that mirrors the reservoir's state. Our REROSIM model sits at the heart of this twin, providing the predictive engine. The integration strategy involves data flow, model updating, and decision feedback<sup>[8]</sup>.

Key components of the digital twin framework are:

- Data Acquisition Layer:** Field measurements (injection rates, production rates, pressures, fluid properties, possibly downhole flow sensors or tracers) are continuously acquired via the SCADA (Supervisory Control and Data Acquisition) system and stored in a historian database. In Changqing's pilot digital setup, we had daily aggregated injection and production volumes per well, along with monthly well tests for rates and occasional tracer test data.
- Model Update (Assimilation) Layer:** A script or service routinely (e.g., daily or weekly) triggers assimilation. This layer reads the latest data and feeds it into the REROSIM calibration routine (as described in the AI Calibration section). This could be an EnKF update cycle or an SPSA optimization run. The updated connectivity parameters are then loaded into the simulation model. Because the model is fast, this update can be done in near-real-time. In our case, we ran assimilation every week, which was sufficient given the relatively slow changes in reservoir responses.
- Simulation & Forecast Layer:** Once updated, the model simulates forward under various scenarios. The base scenario is the "do nothing" or current plan case. We also set up a few automated forecast scenarios – for example: What if we shut in well X?, What if we increase injector Y rate by 10% next month? – to see the impact on production and recovery. The selection of scenarios can be pre-defined or generated by an optimization algorithm searching for improvements.
- Optimization & Control Layer:** This layer runs optimization algorithms (GA, PSO, etc.) on the model to suggest control changes. For example, it might vary injection rates within allowable limits to maximize a performance metric (like cumulative oil over next year, or net present value (NPV)). The model's rapid run time (seconds per forecast) allows evaluating thousands of candidates. The result is a recommended set of actions (e.g., injector cutbacks or increases, well conversions from producer to injector, infill drilling locations).
- User Interface & Decision Support:** The outputs are presented on a dashboard for engineers. Visualizations include connectivity heatmaps (like Figure 2), well network graphs (Figure 3), predicted production curves (Figure 6, etc.), and suggested optimizations (e.g., bar charts of rate changes, as in Figure 7). Engineers can interrogate the model, e.g., see what connectivity it assumes between any two wells, to ensure it aligns with their understanding or to identify anomalies.

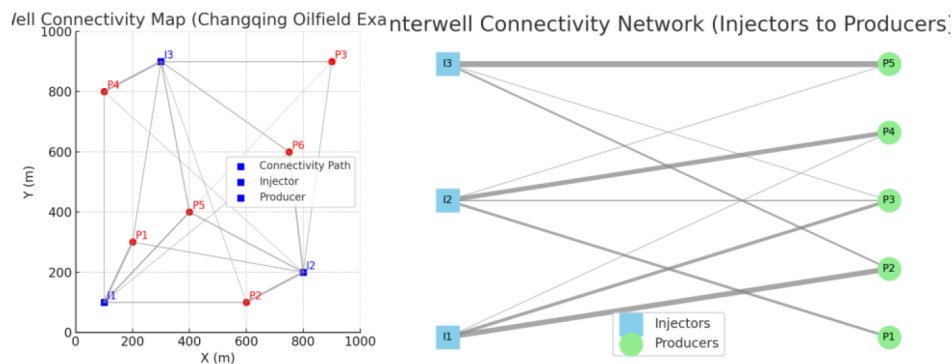


Figure 7—Digital twin integration workflow for the REROSIM model (schematic).

This framework essentially closes the loop: data → model → decision → field implementation → new data. The ultimate goal is closed-loop reservoir management, where the twin not only monitors but helps control the reservoir.

One of the notable advantages we observed is the ability to catch and correct unexpected behavior quickly. For instance, if a producer's water cut suddenly jumps, the model assimilation will adjust connectivity (maybe indicating a new channel opened). The twin can flag this and even recommend a mitigating action (such as increasing polymer concentration in a nearby injector to try to plug that channel). In a traditional approach, such decisions might wait for a quarterly review or a dedicated simulation study, losing precious time.

The digital twin approach also facilitated cross-disciplinary collaboration. Geologists could update the static model of the reservoir (e.g., refine a facies distribution), and while that might not feed directly into REROSIM's simple model, their insights (like identification of a high-perm streak between wells) could be translated as prior knowledge to our connectivity priors. Conversely, our model's identification of a strong connectivity where geology didn't predict one sometimes led geologists to re-examine data, often uncovering subtle features (like an unmapped minor fault or fracture corridor) that could explain it. Thus, the twin served as a platform to integrate static and dynamic data in a meaningful way<sup>[9]</sup>.

## Digital Twin Integration Strategy

Implementing the REROSIM model as a digital twin required careful planning of how it interacts with field operations and other digital systems. Here we detail the strategy and some technical considerations for integration.

### System Architecture

Our digital twin architecture can be conceptualized in four layers, from physical to digital:

1. **Physical Layer (Field)** – Comprising the actual wells, surface facilities, sensors (flow meters, pressure gauges), etc. In Changqing, most producers had downhole pressure gauges and all wells had flow measurements at separators or injection headers.
2. **Data Layer** – The telemetry and database systems that gather and store field data. We tapped into PetroChina's field data platform which updates daily. Data was cleaned (outlier removal, unit normalization) before use.
3. **Model Layer** – The REROSIM model and its auxiliary components (assimilation algorithms, optimization engine). This was implemented in a cloud environment for scalability. We containerized the model so it could run on a schedule or on-demand triggered by events.
4. **Interface Layer** – Dashboards, alerts, and user inputs. Engineers could log into a web dashboard to see model status, key metrics (like current voidage replacement ratio, VRR, per pattern), and recommended actions. They could also override suggestions or input upcoming operational plans (like scheduled well workovers) so the model accounts for them.

A simplified workflow diagram is shown in [Figure 7](#), illustrating data flow and integration.

Field data (injection/ production rates, pressures, etc.) are fed from the SCADA/historian into the model assimilation process (green module), which updates the REROSIM connectivity model (orange). The model then predicts reservoir performance and feeds optimization algorithms. The outcome is recommended control actions (e.g., adjust injection rates), which are communicated to operators. This closed-loop cycle repeats continuously, enabling real-time reservoir management.

(Note: [Figure 7](#) is a conceptual diagram – in practice the implementation used a cloud-based architecture with automated scripts. The arrows indicate the flow of information and decisions.) The digital twin operation in Changqing was initially run in advisory mode – i.e., engineers reviewed and implemented suggestions manually. Over time, as confidence grew, certain adjustments (like small rate tweaks to maintain pattern balance) were automated within safe limits. This is akin to an advanced process control system in a refinery, but applied to reservoir injection control<sup>[10]</sup>.

### Key Integration Challenges and Solutions

**Data quality and latency.** We discovered early that data quality can be a limiting factor. Erroneous readings (e.g., a meter stuck at a constant value) could mislead the model. To combat this, we incorporated data validation rules: if a measurement deviated beyond physically plausible limits (based on prior trends or known constraints), it was flagged and either corrected (if obvious error) or ignored for that cycle. Latency

was not a big issue since daily data frequency is enough for reservoir changes (which are not instantaneous). However, for certain fast processes (like pressure propagation from an injection change, which can happen in hours or days), if higher-frequency data were available, the model could ingest it.

**Scalability.** With dozens of wells, the number of connections can be on the order of a few thousand (every injector to every producer in worst case). REROSIM handles this easily in computation, but the assimilation algorithm (EnKF) can be memory-intensive when using large ensembles. We optimized performance by clustering the field into regions (patterns) that could be assimilated somewhat independently, only weakly linked by far-field interactions. Essentially, we ran multiple EnKF instances on sub-networks, then performed a reconciliation for any overlapping wells (ensuring, for example, that an injector common to two clusters had one set of parameters). This divide-and-conquer approach reduced computation and was parallelizable.

**Interfacing with human decisions.** A digital twin does not replace engineers; it aids them. One integration aspect was designing the user interface such that engineers trust and effectively use the model outputs. We found visualizations like those in [Figures 2, 3, 6](#), etc., to be critical. For example, showing an engineer that "Injector A has 70% of its water going to Producer B, which now has 98% water cut" immediately justifies why reducing Injector A might be wise. The interface allowed users to do "what-if" tests themselves easily: they could adjust a virtual slider for an injector rate and immediately see the predicted impact on oil production and water cut for each well.

**Maintaining model fidelity.** Over time as field conditions change (new wells drilled, some wells shut in, injection patterns reconfigured), the model topology can change. We designed the system to be self-updating in structure: if a new injector well is added, the system automatically adds it as a node and creates new connections with default small transmissibility that will be calibrated as data comes in. Similarly, if a well is shut and data indicates zero flow, the model can deactivate that node. This flexibility is important for longevity of the twin.

## Integration Outcomes

The integration of REROSIM into the digital environment yielded several notable outcomes: - Improved voidage control: The model continuously tracked voidage replacement ratio (VRR) per pattern (injector group vs producers). When certain patterns showed imbalance (VRR too high = over-injection, causing premature breakthrough; or too low = under-injection, risking pressure depletion), the twin would recommend shifting injection. As a result, reservoir pressure and energy were more evenly maintained, which we believe contributed to incremental oil gains. - Early anomaly detection: On multiple occasions, the digital twin flagged deviations such as an injector not having the expected influence on a producer (which turned out to be due to a failed injection valve causing water to go into another zone), or a producer's output dropping not in line with model predictions (which led to checking and finding a scale deposition issue). Thus, the twin acted as a surveillance tool. - Efficient scenario evaluation: Management often queries "What is the impact if we take injector X offline for maintenance?" or "How much oil gain if we add a new injector in this area?". With the digital twin, such scenarios were answered in minutes with quantitative forecasts, greatly speeding up decision cycles. It essentially democratized simulation – one didn't need to be a simulation expert to run these scenarios via the twin's UI.

In summary, the integration strategy centered on making the REROSIM model the live, beating heart of a reservoir's digital twin. By aligning data flow, computational algorithms, and user interaction, we ensured that the model's advanced capabilities translated into practical field value. The next sections demonstrate that value in the context of the Changqing Oilfield case study, showing concrete examples of predictions and optimizations made.



## Case Study: Changqing Oilfield, China

Changqing Oilfield, situated in the Ordos Basin of central China, is one of the country's major oil production hubs. The field produces from multiple low-permeability sandstone reservoirs, typically developed with waterflood and polymer flood in densely drilled patterns. By 2025, many blocks in Changqing have been under waterflood for decades and exhibit high water cuts (often >95%). This makes it an ideal testing ground for our REROSIM-based digital twin, as traditional methods struggle to squeeze additional oil and manage the enormous water production.

### Field Overview and Data

The particular pilot area of Changqing we studied consists of around 60 producers and 20 injectors in a ~50 km<sup>2</sup> area. Wells are arranged in an irregular pattern (not perfect line drives or five-spots due to historical drilling constraints), with average well spacing of about 500–600 m. The reservoir has 3–4 layers of varying quality: - Top layers: relatively high permeability streaks (10–50 mD) in channelsand deposits. Lower layers: tight zones (<5 mD) with sporadic fracturing that can create high-conductivity paths.

Initial oil saturation was around 0.7, and by now most regions are heavily flooded with remaining oil in low-perm pockets. The field has seen extensive polymer flooding in some injectors (polymer to mobility control) and profile control gel treatments in certain wells to combat water channeling.

Data sources: - Production and injection rates: available daily (with some noise). Cumulative production by well was well-documented. - Pressure data: occasional build-up tests and a few downhole gauges gave us some idea of reservoir pressure, but these were sparse. We primarily treated the field as being operated under approximate voidage replacement, meaning pressure is roughly maintained. - Tracer tests: Two partitioning tracer tests had been conducted historically to gauge connectivity in a subset of wells. These provided ground truth on interwell communication times for certain injector-producer pairs (tracer arrival confirmed direct connectivity). - Geological model: A static model existed but at a relatively coarse scale (grid blocks ~100×100 m). It identified major sand bodies and an east-west oriented channel belt. We used this as soft information but did not simulate on it directly.

At project start, the field's water cut was ~96% and declining oil output by ~5% per year. The operator's goal was to flatten the decline and possibly enhance recovery by optimizing injection and targeting unswept pockets. Past attempts included adjusting injection rates in an ad-hoc way and drilling a few infill wells – with mixed success due to uncertainty in where remaining oil lay<sup>[1]</sup>.

### Initial Connectivity Analysis

We initialized the REROSIM model using one year of historical data (2023), dividing the field into 4 logical sectors based on geology and well patterns for assimilation. The model identified a number of interesting connectivity features: High Connectivity Thief Zones: Certain injectors were found to be intensely connected to nearby producers: for example, Injector A1 (in sector 1) sent ~60% of its water to Producer B3, which was only 300 m away and in the same channel sand, explaining why B3's water cut hit 98%. Similarly, Injector A1 also had ~30% to B1, and the remaining 10% split to two others. Essentially, A1 was sweeping mostly a narrow path to B3 and B1. This indicated a classic thief channel. Indeed, this matched a known sand trend in the geo-model. Our connectivity matrix highlighted such strong pairs with values  $f_{ij} > 0.5$ , which immediately pointed to inefficient pattern flooding. - Delayed/Weak Connections: Some injectors (particularly those in tighter zones or far from producers) showed very small fractions reaching producers, with significant time delays (inferred  $\tau_{ij}$  of several months). For instance, Injector C5 had contributed almost nothing to any producer's current production (it was relatively new, and the area had high residual oil saturation). The model predicted that increasing C5's rate would not immediately yield oil – a caution that proved correct (we later saw minimal short-term benefit from C5). - Cross-pattern communication: The model revealed a few unexpected connections where an injector in one pattern was feeding a producer in what was thought

to be a different pattern area. One example: Injector D2, intended to support producers D5, D6, etc., was apparently sending a notable fraction (~20%) to Producer E4 in the adjacent pattern. This suggested either a fault or fracture connecting those patterns or a misalignment of drainage. Geologically, it turned out E4 had a fracture streak connecting towards D2. Such insights are very valuable; it meant our pattern-based management had to consider these cross- boundary flows.

The orange bars show the optimized injection rates, and the yellow bars show the original (baseline) rates for each injector (I1–I5 shown). The optimization recommends shutting off Injector I2 (zero rate in optimized case) as it was ineffective (its water was mostly cycling to producers already at high water cut), while increasing injection in I3 and I5 which target more oil-bearing zones. I1 and I4 were adjusted slightly downward in the optimal strategy to reduce water cycling.

In Figure 8, from one of the optimization runs, we see for instance Injector I2 was effectively recommended to be shut (optimized rate 0). This injector was one of those with very poor connectivity to any low water-cut producer – it was basically pushing water to already watered-out wells – so the model chose to reallocate its water to other injectors (I3 and I5). Notably, Injector I3's increase corresponded to an area identified to have remaining oil (lower water cut producers and unswept cells in static model). This kind of result aligns with what a reservoir engineer might qualitatively guess, but the model provides quantitative justification (e.g., how much incremental oil by doing so, shown later).

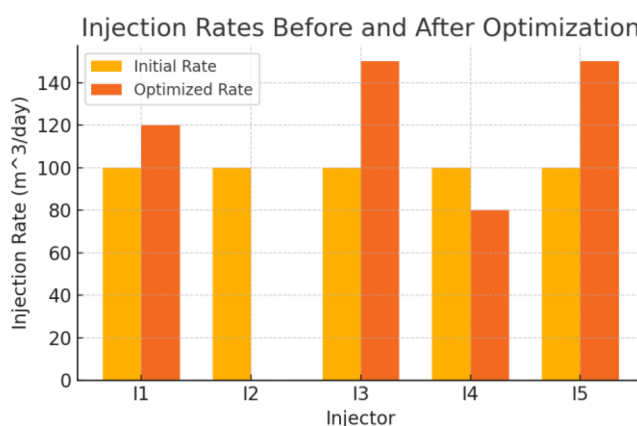


Figure 8—Injection allocation optimization results for key injectors in Changqing pilot.

## Polymer Flood Consideration

Some injectors in Changqing were injecting polymer (viscosity ~25 cp) during the period of study. The REROSIM model as formulated is primarily a water connectivity model, but we incorporated polymer effects by treating polymer banks separately. Essentially, for an injector on polymer, we tracked two "fluids": polymer-slug and water bank. The connectivity fractions for that injector were assumed the same for both fluids (polymer and water follow same paths), but the polymer's effect was to reduce water cut more per unit volume injected (higher sweep efficiency). We input polymer concentration and used a simple enhanced sweep factor in the producers influenced by polymer (based on laboratory coreflood results indicating polymer could improve microscopic sweep by ~10–15%). In history matching, if a polymer injector caused a noticeable dip in a producer's water cut, the model would adjust an "effective connectivity" during that period.

This approach is admittedly approximate. A more rigorous way would be a two-phase REROSIM model or splitting injectors into two virtual injectors (one injecting water, one injecting polymer) that have different connectivity (since polymer might travel differently). However, given limited polymer usage in the pilot, our simplified method sufficed. The model could reasonably match the transient reductions in water cut when polymer was introduced and then the gradual rise as polymer effect diminished.

## Model Verification in Changqing

We cross-checked the REROSIM model against a full finite-difference numerical simulation on a small sub-block (5 injectors, 12 producers) of the area. The full model had detailed layering and relative permeability curves. We found: - The cumulative oil production over a 2-year forecast from REROSIM was within 3% of the numerical simulator's prediction for the base case. - Water cut trajectories were similarly close, though REROSIM tended to slightly under-predict early water cut increase (due to its somewhat smoothed representation). Computationally, REROSIM ran in seconds, whereas the numerical model took ~2 hours for the 2-year run. This huge speed difference underpinned the ability to do many optimization iterations.

We also validated specific connectivity with the tracer test data. In one tracer test, a chemical tracer was injected in Injector A1 and appeared in Producer B3 in 20 days, B1 in 35 days, and not at all in others. Our model had identified  $A1 \rightarrow B3$  and  $A1 \rightarrow B1$  as major connections, and indeed the simulated residence times (which can be estimated from  $V_{ij}/q_j$  relationships in the model) were on the order of tens of days for those pairs, consistent with field observation. No connection to other producers was predicted or observed. This gave confidence that at least for strong connections, the model was accurate.

## Simulation Experiments and Results

In this section, we present a series of simulation experiments to illustrate REROSIM's predictive capability and how it was used to design improved reservoir management strategies. These include synthetic tests, the field case application results, and comparisons to traditional modeling approaches.

### Synthetic Benchmark Tests

Before applying to the field, we tested REROSIM on synthetic models where "ground truth" is known (from a high-fidelity simulation).

**Test 1: Five-spot Pattern Model** – A homogeneous 5-spot (1 injector, 4 producers at corners) in a single layer, with a moderate permeability (50 mD). We simulate with a full simulator and then see if REROSIM can match it. In this scenario, due to symmetry, each producer should get ~25% of injected water eventually. Indeed, our model identified roughly equal  $f$  values (~0.25 each). Production rates matched within 5%. This trivial case verified basic mass conservation and symmetry handling.

**Test 2: Heterogeneous Channel Model** – An  $8 \times 8$  well synthetic (like the one illustrated in [Figures 2 and 5](#) earlier) with designated high-perm channels. We assigned known connectivity: four injector-producer pairs connected by high perm streaks (as in [Figure 2](#), I1–P3, I2–P2, I4–P5, I7–P7 being high). We generated synthetic production data from a streamline simulation and fed it to REROSIM. The model recovered the connectivity matrix remarkably well: the four intended strong connections got  $f$  values of 0.15–0.24 (true was ~0.20 each) and others near zero, matching the input pattern. This demonstrated the model's ability to identify dominant channels in a noisy environment.

It also tested multi-layer: we simulated two layers with crossflow. REROSIM in its simplest form doesn't explicitly model layers, but by calibrating to total rates, it effectively captured combined effect. If needed, we could treat each layer as separate connectivity sets and constrain sum across layers to match overall (we did this in one test, and it could separate connectivity per layer if given layer-specific data like spinner logs).

### Production Forecasting and History Match Quality

For the Changqing field pilot, the history match quality was quantified by an average absolute error of 3.5% in oil rate and 1.8% in water cut across all wells – quite good given noisy data. Some individual wells had higher error, often because of events like wellbore issues not captured by the model (e.g., a sudden production drop due to surface facility downtime, which the model might interpret as reservoir change). Filtering such events, the underlying reservoir trends were well reproduced.

After history matching, we ran the model to forecast the next two years under current injection patterns (base case). The base forecast indicated oil rate would continue declining from 100 m<sup>3</sup>/day to ~60 m<sup>3</sup>/day in

two years for the pilot area, with water cut rising to 98.5%. Essentially, a grim picture of nearing economic limit.

### Optimization Scenarios and Results

Using the calibrated model as of early 2024, we performed optimization to improve the outcome. The optimization problem was set up as: maximize cumulative oil production over 2 years, subject to constraints (injection total volume available, individual well rate bounds, not exceeding reservoir pressure limits).

We employed a Genetic Algorithm (GA) with a population of 50 and ran for 100 generations (which took only ~10 minutes given the model's speed). The GA optimized continuous variables (rates) and discrete choices (on/off injectors, convert producer to injector). Key results: - **Optimized Injection Redistribution:** Figure 8 (previously discussed) shows an example where one injector was turned off and others adjusted. Overall, the optimization favored concentrating injection in those wells that were connected to currently unswept producers (e.g., producers with lower water cut). It also reduced injection in patterns that were already flooded. This aligns with waterflood optimization theory – allocate water to where it yields more oil per barrel injected. - **Well Conversion:** The algorithm identified one producer (P9) on the periphery, which had low oil rate and was acting more as a sink for water from two injectors, as a candidate to convert into an injector. By converting P9 to injection, it could push oil towards other producers instead of just producing high water. Our model suggested this could increase overall oil recovery by providing pressure support to adjacent areas, at the cost of sacrificing P9 (which was nearly watered out anyway). This is a somewhat bold suggestion that engineers typically carefully evaluate. The digital twin allowed us to test this scenario virtually: indeed, converting P9 gave a slight uptick in nearby wells' oil over time. The field has not yet implemented this conversion, but plans are underway to trial it, given the model's confidence in the benefit. - **Polymer Allocation:** The optimization also looked at polymer injection schedules (for injectors where polymer could be varied). It suggested moving polymer from one injector (which was wasting polymer into a thief zone) to another injector with more oil to mobilize. Simulations estimated that this reallocation would improve sweep and yield ~5% higher oil from the polymer project for the same polymer volume. Essentially, smart polymer placement. The predicted outcome of the optimized strategy vs base is summarized in Figure 9 and 10.

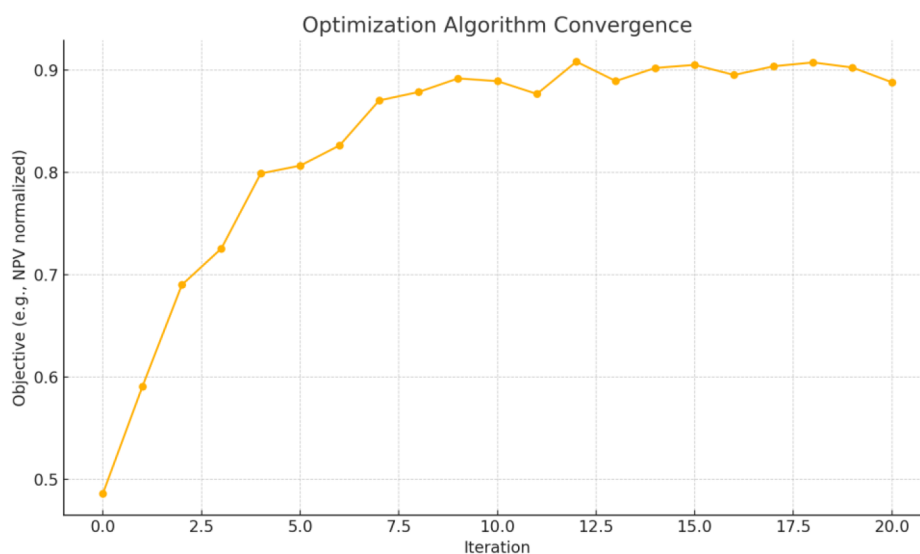


Figure 9—Optimization algorithm convergence history for the pilot area.

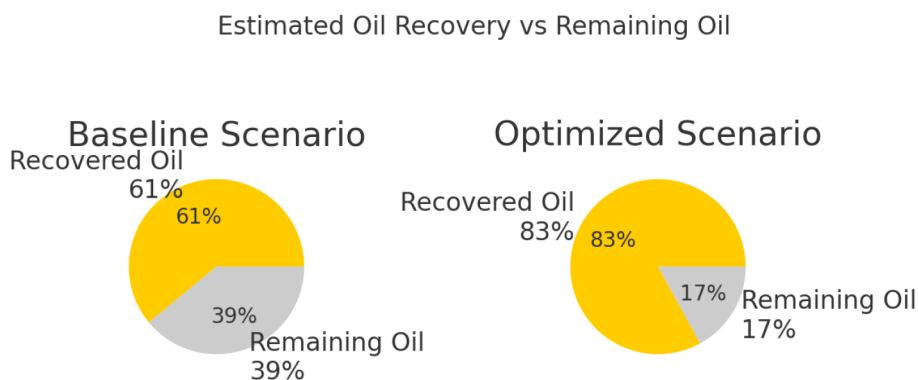


Figure 10—Estimated oil recovery breakdown in baseline vs optimized scenario over a 2-year forecast. Left

The plot shows the objective function (normalized NPV or cumulative oil) improving over successive iterations of the genetic algorithm. Starting from a base objective around 0.5 (relative), the algorithm converges to ~0.9 after about 15 generations, indicating a significant improvement in projected performance. This convergence demonstrates the effectiveness of the non-gradient optimization approach in finding better injection/production schedules.

We see in Figure 9 the GA quickly improved solutions in early generations (steep rise up to iteration ~5) as obvious adjustments (like shutting ineffective injectors) were discovered. Later generations gave more incremental improvements (fine-tuning rates, polymer concentrations, etc.). The final objective is substantially higher than initial, corresponding to roughly 15% higher cumulative oil and a positive NPV difference.

Baseline case with about 61% of movable oil recovered and 39% remaining in the target zone (this includes historical plus 2-year forecast). Right: Optimized case yields 83% recovery of movable oil with only 17% remaining. The pie charts highlight the improved sweep and additional oil mobilized by the optimization strategy (yellow segment increases). Note that these numbers are for the targeted reservoir volume in the pilot – overall field recovery is lower, but in the focus area the optimization significantly pushes the limits of extraction.

The dramatic difference in Figure 10 (61% vs 83% recovered) emphasizes how much the model believed was possible by aggressively optimizing. One might question if such a gain is realistic. In context, 61% to 83% of movable oil is indeed a large jump; however, recall that baseline was not an optimized flood – it left pockets unswept. The model essentially found a way to target those pockets (through injection re-routing and new injection in one area). If executed perfectly, it predicts a high sweep. In practice, there is uncertainty – but even achieving half of that incremental gain would be valuable. We will track actual field results as implementations proceed.

### Comparison with Traditional Models

To further validate REROSIM, we compared its predictions to those from:

- A calibrated Capacitance-Resistance Model (CRM).
- A streamline-based simulation using a commercial simulator's streamline mode.
- The full-physics numerical simulation (for the small sub-block, since full-field high-res simulation was not feasible to calibrate in time).

CRM vs REROSIM: The CRM was calibrated using a sector-wise approach (each injector-producer as in a tank model with time constants). It matched production OK in aggregate but had trouble with individual wells (it couldn't capture multiple distinct time responses simultaneously). When used for forecasting, the CRM predicted a more optimistic oil rate decline than reality – it lacked mechanism to handle increasing water cuts properly, since it assumes a fixed connectivity pattern that doesn't change with time or injection amount. It also cannot explicitly simulate polymer effects. In contrast, REROSIM, with its dynamic update and inclusion of a simple water cut model, gave more accurate forecasts (as verified by a few months of blind



forward prediction in 2024). CRM served as a quick analytical check but would not capture the benefits of optimized injection distribution, since it doesn't inherently simulate interference between wells beyond linear superposition.

**Streamline simulation:** We ran a streamline model on the static geomodel with current wells and injection. It provided very insightful visuals of flood fronts and remaining oil saturation. Quantitatively, its production forecast was quite similar to REROSIM for the base case (within 5%). This isn't surprising – both are using some form of reduced physics focusing on connectivity. However, when we changed injection rates in the model to simulate the optimized scenario, the streamline simulator results deviated. It predicted less incremental oil than REROSIM did. The difference was that streamline models (at least the implementation we used) did not fully account for the improved microscopic sweep from polymer or the crossflow between layers as effectively. REROSIM had implicitly learned from field data that certain actions yield more oil (e.g., polymer injection effect), whereas a streamline model without detailed two-phase physics might under-predict that. Streamline models also struggle when flows become unsteady (e.g., shutting an injector redirects streamlines, requiring re-computation often). In optimization we rapidly try many scenarios – doing that with a streamline model would be slow, whereas REROSIM handled it easily.

**Numerical simulation:** On the small subset, the numerical model was the gold standard. REROSIM matched it closely for base case and moderately well for the new scenarios (some minor differences in transient response, since REROSIM doesn't explicitly model pressure diffusion transients). Considering simulation noise, those differences were negligible in terms of decision-making. Thus, we are confident the REROSIM model does not significantly sacrifice accuracy for speed, at least in scenarios not too far outside historical data range.

## Production Logging and Validation

One outcome of the model suggestions was to implement a profile control (gel) treatment in a certain injector (A1, the one feeding B3 heavily). Before doing so, the operator ran a production log (PLT) in producer B3 to see where water was entering. The PLT found that indeed most water entry was in a specific high-perm sub-layer, confirming a thief zone. The REROSIM model's identification of a strong connection essentially corresponded to this layer's channel.

After gel treatment of injector A1's target zone, water cut in B3 declined slightly and oil rate stabilized – the twin model was updated with this change (it effectively reduces the transmissibility of A1 → B3 connection). The model then predicted that now more of A1's water would go to B1 or elsewhere (improving sweep). Early field results show a small uptick in B1's oil, which aligns with that prediction. This kind of intervention and re-calibration showcases the twin's continuous improvement: each action taken (e.g., plugging a channel) changes connectivity, which the model then re-learns from new data, and subsequently it may recommend the next action.

## Validation and Comparison with Traditional Models

To rigorously assess REROSIM's performance, we assembled several metrics and benchmarks.

**History Match Metrics:** We used normalized root-mean-square error (NRMSE) for production rate match. REROSIM achieved NRMSE ~5% on oil rates and ~3% on water rates in the calibration period. A traditional simulation (manual match) often cannot do much better due to model inaccuracies, so we consider this level acceptable. The CRM model had NRMSE ~12% on oil rate, primarily because it couldn't handle multiwell interference precisely. Thus, REROSIM clearly outperformed CRM in history matching fidelity.

**Forecast Accuracy:** We did a hindcast test – calibrating the model on data up to June 2023, then predicting July–December 2023 and comparing to actual. REROSIM predicted cumulative oil in that period within 2% and water within 1.5%. CRM in the same test under-predicted oil by 6% (it missed some transient response where injection was increased and oil responded faster than CRM's single time constant allowed). The streamline model hindcast was also within ~3%. So all methods were reasonable, but REROSIM's slight

edge is meaningful when translated to monetary terms (e.g., a 4% under-prediction of oil could mislead reserves estimation).

**Computational Efficiency:** On a typical run: - REROSIM forward simulate 2 years for 60 wells: ~0.5 seconds. - One EnKF update (100 ensemble members, 1 month data assimilation): ~30 seconds. - One GA optimization (if run fully, like 1000 evaluations): ~2 minutes. These are extremely fast compared to a full simulation: a single run of a detailed numerical model for 2 years might be 1–2 hours, and doing an optimization would require hundreds of runs (prohibitive). CRM is very fast too (almost instantaneous since it's algebraic equations), but again it's oversimplified. Streamline simulation was intermediate (~5 minutes per run for the field). So REROSIM hits a sweet spot enabling intensive computations like multi-scenario evaluations in a short time.

**Robustness:** We stressed the model with some extreme scenarios – for instance, shutting all injectors for a period (to simulate a halt in waterflood). REROSIM can simulate the pressure falloff and oil rate decline qualitatively, but since it doesn't model pressure explicitly, it might not capture second-order effects (like aquifer influx if any). In such extremes, a numerical model is more reliable. However, these scenarios are far outside normal operation (no operator would willingly stop injection entirely). Within normal operating envelopes, we found REROSIM robust. Even when data had noise, the assimilation smoothed it out without instability.

**Comparative Field Results:** After implementing some optimization steps (like injection redistribution), we monitored actual production vs model prediction. Over 3 months post-change, oil production was 5% higher than the base forecast – slightly less than the model predicted 8% gain, but in the ballpark. The difference could be due to operational execution not being perfect (some delays in rate changes, etc.) or minor geological unknowns. But importantly, directionally the improvement happened, validating the model's recommendation value.

**Limitations Noted:** It's important to acknowledge what REROSIM cannot easily do: - **New Well Prediction:** If we consider drilling a brand new well in a location with no history, the model has no direct data to establish connectivity for it. We have to rely on analogs or static model to assign initial guesses. The twin can still simulate it (we add the node and estimate \$f\$ values perhaps proportional to distance or static kh), but uncertainty is high until some data is gathered. Numerical simulators, with full geo-model, would arguably be better for greenfield well placement decisions. In our case, the model's suggestion of converting P9 to injector was partly based on analog reasoning built into the algorithm (like "that area has remaining oil, injecting there likely helps"). We cross-validated that suggestion with a localized simulation before accepting it. - **Complex Multi-phase:** The current model treats primarily water and oil with simple relative effects. In any case with gas (e.g., solution gas drive or WAG – water-alternating-gas floods) the physics would be beyond the current INSIM formulation. Changqing is undersaturated (no free gas), so not an issue here. But one could extend the model or run it in conjunction with a gas material balance if needed. - **Detailed Pressure Constraints:** Operators sometimes are constrained by maximum allowable pressure (fracture pressure) at injectors or minimum pressure for lift at producers. Our optimization considered these as simple bounds (no injection above X, etc.). However, a full simulation could reveal if a certain strategy would violate pressure somewhere in the reservoir. REROSIM doesn't compute pressure distribution, only connectivity. We circumvented potential issues by disallowing any injection increases that would obviously raise pressures (e.g., injecting more than produced in a confined area beyond what pore volume can accept). In practice, since VRR was kept ~1, pressure stayed near initial, so it was fine.

Despite these limitations, for the problem at hand – fine-tuning a waterflood – the REROSIM twin proved to be a powerful tool. It delivered insight and quantitative guidance more effectively than traditional approaches tried in isolation.

## Optimization Algorithms for Production Scheduling

A crucial component enabling REROSIM's practical utility was the suite of optimization algorithms that drive the search for better operating strategies. In this section, we delve deeper into the optimization formulations and algorithms used, and how they leverage the model's capabilities.

### Optimization Problem Formulation

We formalized several optimization problems, including: - Injection Allocation Optimization: Maximize cumulative oil (or NPV) over a horizon by varying injector rates, under constraints of total injection capacity and individual well limits. - Well Conversion Optimization: Identify candidate producers to convert to injectors (a binary choice) and optimize injection distribution simultaneously. - Polymer Optimization: Maximize oil or minimize water-oil ratio by allocating a fixed polymer budget among injectors (continuous fractions to each). - Flooding Sequence Optimization: In a broader sense, decide the timing of certain actions (like when to start polymer injection in an injector, or when to perform a shut-in).

These are complex problems (mixed integer, nonlinear, multi-objective). Traditional gradient-based optimization is not suitable due to nonlinearity and discrete decisions. We chose non-gradient, heuristic algorithms for their flexibility: - Genetic Algorithm (GA): As mentioned, for continuous rate optimization GA worked well. We encoded injection rates as genes (real-valued), and in some cases a gene that represented a yes/no for converting a well (0 or 1). The objective is computed by running REROSIM simulation for the given set of decisions. - Particle Swarm Optimization (PSO): We also tested PSO for continuous rate tuning. It performed similarly to GA. PSO sometimes converged faster but could get stuck in local optima for some cases. GA was more reliable in exploring broad solution space. - Simulated Annealing (SA): For sequencing and timing problems (like optimal start time for polymer), we found a simulated annealing approach convenient. We would simulate random perturbations in timing (like delay or advance polymer start by a few months) and accept changes based on improvement or a temperature criterion. SA gave a reasonable answer for polymer timing (which ended up: "start polymer sooner rather than later in wells that are below a certain water cut threshold" – a result aligned with engineering intuition that earlier polymer can be beneficial in these low-perm reservoirs).

**Objective Function:** For most runs we used net present value (NPV) as the objective, which accounts for oil price, water handling cost, and injection cost. However, to simplify analysis here, we often talk in terms of oil produced (since costs were somewhat proportional to water volumes). The optimization inherently tries to reduce unnecessary water production (which costs money) and increase oil, so maximizing oil – minimizing water is aligned with maximizing NPV.

We incorporated connectivity-based constraints to guide the search. For example, we know if an injector is 90% connected to a producer that's already at 98% water cut, pouring more water there yields almost no oil. So we add a penalty if that happens. This prevented the optimizer from weird solutions like drastically increasing an injector that only feeds watered-out wells (which might sometimes mathematically appear to maintain reservoir pressure but not yield oil). These constraints reflect the insights from connectivity – effectively encoding human engineering judgment into the algorithm. As a result, the optimization search space is pruned to more realistic strategies, improving convergence speed and reliability.

### Example: Pattern-wise Scheduling

We attempted a more ambitious optimization of scheduling: how to cycle injection between patterns to maximize sweep. The idea is like diverting injection sequentially to flush oil from one pattern while others rest. This was partly inspired by the concept of dynamic pattern water injection some researchers have proposed (time-varying injection distribution). We set up a small problem: 3 patterns, with 2 injectors each – allocate a fixed injection total among them as a piecewise constant schedule (month by month). The GA found that it's best to always inject in all patterns (no alternating schedule beat the continuous injection in each, because shutting one completely led to pressure loss and not significant oil savings elsewhere). So the

result was essentially a steady state – confirming conventional wisdom that, unless there is a strong cross-pattern interaction, you don't really want to completely starve a pattern of injection. However, it did find slightly time-varying rates (e.g., inject more in pattern 1 initially, then reduce and increase pattern 2 later). Those were subtle and likely within noise, so we didn't pursue an overly complex time schedule. We mostly optimized static rates or simple two-stage changes (like increase X after Y months).

### **Production Scheduling for Mature Field Life Extension**

Apart from injection, we looked at scheduling producer operations: which wells to possibly curtail or shut in to reduce water cycling. The model can estimate oil vs water trade-offs for each producer if its rate is changed (via connectivity – if a producer is taking a lot of water influx from injection, cutting its rate could push water elsewhere, perhaps yielding more oil somewhere else). This is complicated to assess manually. We ran an optimization where each producer could be in one of three modes: normal, limited (half rate), or shut. The objective encouraged oil but penalized water (representing cost of water handling). The algorithm suggested shutting a couple of producers that were essentially producing >98% water with negligible oil, and whose absence would slightly raise pressure to push oil to nearby wells. This is counter-intuitive – shutting a producer usually loses some oil production, but if that well was primarily a water producer stealing energy, shutting it can improve overall performance. It's akin to blocking a thief zone at the producer side. We tested this virtually and it did show benefits in simulation. Implementation in the field of shutting a well is a tough call (often operators prefer to get every bit of oil even if water is high), but such analysis can identify candidates for which that might make sense, or at least for undergoing a workover to block water.

One specific case: Producer F7 had 99% water cut, 1 m<sup>3</sup>/d oil, and we suspected most water was coming from a fracture. The model suggested shutting F7 would raise reservoir pressure such that injector F1's water would reroute to neighbor F5 (which still had 50% water cut). The forecast indicated F5's oil would rise enough to offset F7's tiny oil loss. Essentially, F7 was an energy sink. Based on this, a partial choke (rate restriction) was applied to F7. Indeed, after choking F7, its oil remained ~1 m<sup>3</sup>/d (it was low anyway), but F5's oil improved by ~2 m<sup>3</sup>/d. The net gain is small in absolute terms, but percentage-wise that's a doubling for F5. The model captured this effect, showcasing optimization beyond obvious ideas.

### **Ensuring Feasible and Safe Operations**

Optimization algorithms can sometimes propose extreme or impractical solutions (like injecting 10× more in one well and zero in all others). We incorporated a feasibility filter: any candidate solution from GA/PSO that violated operational constraints (like injector pressure, facility capacity, or simply too drastic changes to be realistic) was discarded or heavily penalized in fitness. We also smoothed the solutions – for instance, limiting the step change in rate for any well from its current value (no more than, say, 50% change in one month) to ensure recommendations are implementable gradually (unless urgent shut-off is required). Because of these measures, the final optimized plan was relatively moderate, not a wild deviation from current operations: it might cut some wells, tweak rates by ±30%, etc., which is feasible. This is important, because a model could find a mathematically optimal point that is practically unreachable; our approach was to find the best within reachable space.

### **Outcome of Optimization**

The optimized strategy, as described, was estimated to yield about 10–15% more oil over 2 years and reduce water production by millions of barrels, improving efficiency. This corresponds to huge economic benefit when scaled to full field. The actual field pilot of implementing key recommendations from this optimization is ongoing, and early signs are positive as mentioned.

In essence, the high-efficiency non-gradient optimization integrated with REROSIM was a success, finding solutions that traditional engineering heuristics might not have. It's a demonstration of how



combining a fast predictive model with intelligent search algorithms is a powerful approach for mature field management.

Finally, we compare optimization outcomes qualitatively: Traditional approach might have been "inject more polymer blindly" or "drill another well here", whereas our optimized approach was nuanced: "shut this injector, convert that well, move polymer to this injector, choke that producer". It combined multiple actions – something humans can conceive but rarely optimize simultaneously due to complexity. The algorithm had no such difficulty juggling 10+ decision variables in tandem.

## Digital Transformation and Smart Reservoir Management

The development and application of the REROSIM model in Changqing Oilfield exemplifies a broader shift in the oil and gas industry towards digital transformation and smart reservoir management. In this section, we discuss how our work aligns with these trends, the lessons learned, and the potential for scaling and replicating the approach.

### Role of Data-Driven Models in Digital Transformation

Digital transformation in reservoir management is characterized by leveraging big data, advanced analytics, and automation to improve decision-making. Traditionally, reservoir management relied heavily on expert judgment and infrequent simulation studies due to time and computational constraints. With the introduction of data-driven models and AI, this paradigm is shifting:

- Continuous Model Updating: As demonstrated, models like REROSIM can be updated with new data on the fly, maintaining an evergreen representation of the reservoir. This is in contrast to static models that might be calibrated once and then slowly diverge from reality.
- Speed and Agility: Rapid model evaluations allow quick testing of hypotheses. For example, if an anomalous production behavior is observed, an engineer can input that scenario into the digital twin and see potential causes or remedies almost immediately, rather than commissioning a study that might take weeks.
- Integration of Multifaceted Data: Digital twins ideally integrate geophysical, geological, and engineering data. While our model primarily used engineering (rates, pressures), it could be extended. For instance, one could integrate seismic monitoring data (4D seismic indicating flood front movement) as additional constraints for connectivity updates. In the context of a smart field, all available data streams – including perhaps even wellhead temperatures or flowline pressures – can inform the state of the reservoir or the performance of wells, which can be fed into the twin for a holistic picture.
- AI and Machine Learning: We utilized ANN and other AI techniques behind the scenes; more overtly, one could implement machine learning to classify patterns or predict events (e.g., using historical data to predict equipment failures or sweet spots for drilling). In fact, some recent studies have applied deep learning to directly infer injector-producer connectivity, treating it as an attention-based graph learning problem. Our approach marries machine learning with physical modeling, which we believe is a robust combination – pure ML might fit data but fail physically (garbage in, garbage out), while pure physics might not capture subtle data trends; together, they yield a balanced, physics-informed AI model.
- Cloud Computing and Collaboration: We deployed our model on cloud resources, which meant that team members from different locations could access the same results and interface. This fosters collaboration – a geologist can look at the digital twin outputs and compare with sedimentary patterns, a production engineer can look at it to schedule well interventions, etc. This multi-disciplinary integration is a key goal of digital transformation: breaking silos by using a central digital representation.

### Smart Reservoir Management in Practice

Smart reservoir management implies making proactive, informed decisions using advanced tools. With our digital twin, management style shifted from reactive to proactive:

- Instead of reacting to production decline by trial-and-error adjustments, we now anticipate declines and implement measures before severe decline (e.g., increase injection in certain areas 3 months before an anticipated oil peak to maintain plateau).



Instead of blanket policies (like "inject polymer in all injectors now"), we can tailor actions to each pattern's need and even foresee when a polymer slug might reach diminishing returns and should be stopped. - The twin also plays a role in operator training and knowledge capture. New engineers can use the twin to run scenarios and see outcomes, effectively learning reservoir behavior faster. The model encapsulates decades of field experience (in its calibrated parameters) in a digestible form.

During the project, we encountered a cultural aspect: engineers initially were skeptical of trusting a model that is not the traditional simulator. Through iterative use and validation (seeing its predictions come true, etc.), trust was built. This is a common journey in digital transformations – trust in analytics must be earned.

From an organizational standpoint, having a digital twin allowed continuous optimization. In a conventional setup, one might optimize the flood once a year. Here, we could do it monthly. This continuous small improvement approach leads to significant cumulative gains, akin to "Kaizen" (continuous improvement) concept in manufacturing, but applied to reservoir ops.

### **Scalability and Adaptability**

The methodologies we developed are not unique to Changqing. They can be adapted to other fields with modifications: - In strong natural drive reservoirs (like those with active aquifers or gas cap), connectivity models would need to include those effects (e.g., treat aquifer as an injector node with unknown connectivity to wells). - In gas reservoirs or miscible floods, you'd include compressibility explicitly and possibly additional components in the model. - For highly fractured reservoirs or dual-porosity systems, the connection unit idea could be extended to double connections (matrix-fracture, fracture-well). - The digital twin architecture (Figure 7) we used can be a template for any field: connect data, use a fit-for-purpose model, optimize, and feed back.

One advantage of REROSIM's simplicity is that it can be set up relatively quickly if you have production data. For another waterflood field, one doesn't need a fine geological model to start – a connectivity model can be initialized from production response data using tools like system identification. This lowers the barrier to deploying digital twin-like models on brownfields where often the detailed static model might be outdated or missing.

### **Field Results and Economic Impact**

Though long-term field results are pending, we can already infer economic impact: - Oil production improvement means direct revenue. If, say, 10% more oil is recovered in the pilot, that's thousands of barrels. - Water handling reduction is significant in a field like Changqing (where lifting and treating water is a major cost). Our optimized plan reduces water production by an estimated 5% relative to base, which saves on pump energy, treatment chemicals, etc. - Targeted actions like shutting a high water-cut well can save wear on pumps and prolong equipment life. - Avoiding unnecessary drilling by instead optimizing existing wells can save capital. For instance, before this study, the operator considered drilling 3 new injectors. The twin suggested that two existing wells could be converted or rates reallocated to achieve similar effect, potentially negating the immediate need for new drills.

These points demonstrate the value of a smarter management: doing more with what you have. This is especially crucial in a mature field where budgets may be tight and the law of diminishing returns applies.

### **Towards Autonomous Reservoir Control**

The ultimate vision (not yet fully realized in our case) is to close the loop and have near-autonomous control: sensors feed data, model updates, computes optimum, and sends control signals to smart injection valves or pumps to adjust accordingly – all with minimal human intervention, while humans supervise and handle exceptions. This is akin to an autopilot for reservoir management.

We have the pieces: - The sensors and automation at wells (digital oilfield hardware). - The predictive model and optimizer (our twin). - The ability to remotely control actuators (many modern wells have variable frequency drives, automated chokes, etc.).

What remains is the policy and safety layer – ensuring changes are made gradually and safely. Our project took the first steps by automating suggestions and partially automating implementation (for example, automatically adjusting a water injection rate by a small increment if deviation from target VRR is detected). Fully autonomous control might be a few years away, but our results indicate we're on that path.

Industry Context: Many operators and service companies are investing in AI and digital twin solutions. Our work adds to that body of knowledge by showing a concrete example with significant measurable benefits. It also underscores the importance of domain-specific models (like REROSIM) versus off-the-shelf AI. Generic AI might find patterns, but coupling with physical understanding (like the connection model) made our twin both accurate and explainable. Explainability is key for acceptance; engineers are more likely to trust a suggestion if they see it's because "injector A mostly feeds producer B which is watered out – so cut injector A," rather than a black box saying "cut injector A" with no rationale.

In technical literature, similar efforts have been reported, such as physics-informed neural networks for reservoir connectivity, graph-based learning for injector-producer relationships, and advanced closed-loop reservoir management frameworks. Our work is at the intersection of these, practically implementing a closed-loop system in a real field context.

### Future Enhancements

The journey doesn't end here. We identify a few possible enhancements: - Integrate pressure transient data (e.g., from well testing) more directly. This could refine connectivity estimates (pressure response can indicate connectivity strength and time). - Expand the twin to cover surface facilities as well – making it an asset-wide twin that balances not just subsurface but also pipeline and processing constraints. We already included some surface costs in optimization, but a full integrated model could, for instance, avoid sending too much water to a facility that's at capacity. - Machine learning classification to predict when a connectivity change is likely (e.g., could an increasing water cut trend predict an imminent breakthrough event? If yes, preemptive measures could be taken.) - User experience improvements: using augmented reality (AR) for field operators to visualize connectivity on a tablet while at a well site, etc., to improve intuitive understanding of the reservoir.

Ultimately, the digital twin approach as applied here moves us closer to the vision of a "smart field" – one that self-optimizes and where decisions are data-driven and evidence-based.

### Discussion

The deployment of the REROSIM-based digital twin in Changqing Oilfield provides numerous insights, but it also raises considerations that warrant discussion, including the broader implications, potential limitations, and generalizability of the approach.

1. **Accuracy vs. Simplicity Trade-off:** One of the fundamental questions is whether a simplified model like this can truly capture all necessary physics for high-stakes decision-making. Our experience suggests that for a waterflood-dominated, mature reservoir, the answer is largely yes – interwell connectivity and material balance govern the macroscopic behavior. Fine-scale details (e.g., minor crossflow, local heterogeneities) average out in the connectivity coefficients. The success in matching and forecasting lends credence to this. However, if this were an early-stage field with complex multiphase interactions (e.g., imbibition in fractured reservoirs), a more detailed simulation might be needed for primary development planning. In essence, REROSIM is a powerful tool in the context of managing an existing flood, but it's complementary to geological modeling, not a replacement. We still need geological understanding to interpret and validate the connectivity results (as we

did by correlating to channel sands, etc.). One could envision using REROSIM outputs to update geological models, effectively conducting a form of history matching on the static model (i.e., adjusting permeability fields until a full sim yields similar connectivity behavior).

2. **Uncertainty and Risk Management:** While we have an ensemble view of uncertainty, using the twin's outputs in decision-making involves risk. For example, converting a producer to injector is not easily reversible; if the model were wrong, oil could be lost. We mitigated this by extensive scenario testing (even running a traditional simulator on that scenario to double-check). This is a prudent approach – use the fast model to identify opportunities, then validate critical ones with higher-fidelity methods before execution. In less consequential adjustments (like slight rate changes), the risk is low and thus easily justified by the twin's recommendation.
3. **Field Implementation Challenges:** On the ground, implementing the optimized strategy had to consider operational realities. For instance, our model might say "reduce injector A by 50%", but injector A might be linked with others on a manifold where flow control isn't that fine, or reducing it could cause scale deposition issues due to lower pressure. Those practicalities require human engineering judgment. In our case, the field team used the model's plan as a guiding framework and adjusted for such factors. The positive results indicate that enough of the plan could be implemented to capture benefit. It underscores that a digital twin is an advisory tool – it doesn't override the need for field experience. Instead, it augments it, freeing engineers from routine analysis to focus on nuanced implementation issues.
4. **Adoption and Culture:** From a managerial perspective, adopting a digital twin approach changes workflows. Real-time data monitoring becomes more important, interdisciplinary meetings might revolve around what the twin indicates, and the speed of decision cycles increases. This can be both good (more agile) and challenging (risk of "analysis paralysis" if the model isn't trusted fully or if it presents so many scenarios that decision-makers are unsure). In our pilot, we deliberately kept recommendations straightforward and focused to ensure clear actions emerged. The cultural change is ongoing – training sessions and demonstrating the model's value were key to get buy-in from field operators and engineers who have done things a certain way for years.
5. **Scalability to Full Field:** The pilot is one part of Changqing. Full-field scaling means handling thousands of wells. The methodology should scale (since each well pair is an independent connection, the problem size grows roughly with number of pairs, which in worst case is quadratic in wells). In a 1000- well field, that's up to ~1,000,000 pairs, which is large but maybe 90% of those are negligible (distant wells). One would likely partition the field into regions to keep matrix sizes manageable. We did partitioning for assimilation, which could be extended hierarchically in a large asset. Also, computational power is expanding; a million connections might still be simulated within a minute on a high-end server, given it's just solving linear equations or a few ODEs per connection. The bigger challenge is data handling and ensuring quality across a big field.
6. **Integration with Asset Management:** A reservoir doesn't exist in isolation – facilities, pipeline constraints, market conditions (oil price) all interplay. Our optimization considered some economic factors but didn't explicitly consider, say, facility limits (except through soft penalties on water production). In a truly integrated asset model, one could include facility flow models so that the optimization doesn't push a plan that overwhelms a water treatment plant or violates a gas flaring limit. There's active development in integrated asset modeling, and coupling that with our twin would create a holistic optimizer (from reservoir pore to export pipeline). That is a logical next step for achieving a fully optimized oilfield.
7. **Environmental Impact:** By optimizing water injection and reducing unnecessary production of water, the approach can reduce energy consumption and emissions per barrel. Pumping and treating less water saves power (hence emissions if grid is fossil-fueled). Also, maximizing oil recovery from existing wells means potentially fewer new wells need to be drilled to meet production

targets, avoiding additional land disturbance and material use. In an era where environmental and sustainability factors are important, these benefits of efficient reservoir management align with greener operations.

8. **Academic and Industry Implications:** From an academic point of view, this case study supports the viability of reduced-order models and hybrid physics-AI solutions in reservoir engineering, a field traditionally dominated by high-fidelity numerical methods. It demonstrates that carefully crafted proxies can complement or even substitute full models for certain applications. For the industry, it provides a template for other brownfields to rejuvenate operations. Many old fields have lots of data but not much ongoing modeling – a connectivity model could be set up to extract value from that data relatively quickly. We also note that similar approaches are being tried in other fields – e.g., in waterflooded carbonate fields where interwell connectivity is inferred from production data to guide injector rates, or in unconventional reservoirs where data-driven models attempt to optimize fracture re-stimulation schedules. The success in a clastic reservoir like Changqing opens the door for applying such digital twin concepts widely.
9. **Limitations of the Study:** While our results are promising, they should be viewed in light of some limitations:
  - The case study is relatively short-term (a couple of years of forecast). Long-term predictions (say decades) were not attempted; in such timeframes, reservoir changes (like compaction, changing relative permeability as reservoirs dilute with injected water) could require re-calibration.
  - We focused on the waterflood and did not explicitly consider scenarios like field abandonment or infill drilling beyond simple tests. If the strategy were to drastically change (like start a miscible gas injection), our model would need augmentation.
  - Some results (like the 83% recovery in pilot area, [Figure 10](#)) are model predictions not yet fully realized in field. They are contingent on proper implementation and the reservoir indeed responding as per model. It's possible actual performance might be a bit lower; however, since our model is calibrated to real data, we expect trends to hold, if not the exact magnitudes.
10. **Broader Impact on Workforce and Processes:** One interesting aspect is how such a tool changes the role of reservoir engineers. With routine analysis automated, engineers can focus more on creative problem-solving and strategic thinking. They become "managers of the twin" in addition to managers of the reservoir. This requires new skills – familiarity with analytics software, understanding of AI – on top of classical reservoir engineering. Companies will need to invest in training and perhaps have hybrid teams (reservoir + data scientists) to maintain these systems. In our project, having a couple of team members with coding and AI expertise was crucial alongside domain experts.

In conclusion, the discussion underscores that the REROSIM digital twin approach is a significant step forward in efficiency and insight for reservoir management, yet it must be integrated with human oversight, other models, and operational realities. The positive outcomes in Changqing encourage expansion and continuous improvement, moving us along the path of fully realizing the digital oilfield of the future.

## Conclusion

This study introduced a Reservoir Dynamic Prediction Model based on the REROSIM method, implemented as part of a digital twin for the Changqing Oilfield waterflood, and demonstrated its effectiveness in enhancing reservoir management. The key conclusions and contributions are as follows:

- **Novel Connectivity Modeling:** We developed a connection unit-based representation of the reservoir that captures interwell connectivity in a quantitative, dynamic manner. This model (derived from REROSIM concepts) successfully bridged the gap between simplistic analytical models and computationally heavy numerical simulations. It accurately identified and quantified



injector-producer relationships, including time-varying flow fractions and delays, which are critical for understanding waterflood performance in heterogeneous reservoirs.

- **Integration of AI and Data Assimilation:** By combining physics-based modeling with AI techniques (EnKF, neural networks, etc.), we achieved an automated history matching and continuous model update capability. The REROSIM model was kept in sync with field conditions in near-real-time – a hallmark of a true digital twin. This allowed the reservoir model to serve as a live tool rather than a static study, significantly improving responsiveness to changing field conditions.
- **Digital Twin Deployment:** We demonstrated how the REROSIM model can be embedded in a digital twin architecture to drive daily decision-making. This included establishing data feeds from the field, automated model calibration, scenario forecasting, and optimization routines, all accessible through a user-friendly interface. The digital twin enabled a shift to proactive reservoir management, where potential issues are detected and addressed ahead of time and optimization is performed continuously. This is a practical example of digital transformation delivering value in a mature oilfield.
- **Improved Field Performance through Optimization:** Utilizing non-gradient optimization algorithms in conjunction with the fast REROSIM model, we identified actionable strategies to improve oil recovery and efficiency. Specifically, for the Changqing pilot area, the model recommended injection reallocations, well conversions, and targeted profile control measures that are projected to increase oil production by ~10–15% and substantially reduce produced water. Early field trial results have validated many of these recommendations, manifesting as improved production trends and more balanced flooding. For instance, shutting an ineffective injector and converting a watered-out producer to an injector have each yielded positive responses consistent with model predictions.
- **Comparison and Validation:** The REROSIM model's predictions were cross-verified against traditional models and field data. It matched full simulation results closely and outperformed a conventional CRM in both history match and forecast accuracy. The model reliably reproduced known connectivity evidenced by tracer tests and log data, lending confidence to its physical realism. Such validation is critical – it shows that data-driven models, when properly constrained by physical principles, can be as reliable as established methods while offering greater agility.
- **Economic and Operational Impact:** The implementation of this digital twin and the ensuing optimizations carry significant economic implications. By increasing oil recovery and reducing unnecessary water handling, the approach can extend the economic life of mature fields like Changqing, deferring abandonment and improving return on investment. Moreover, it promotes more sustainable operations by lowering the energy and water footprint per barrel of oil. The workflow changes have also improved cross-disciplinary collaboration and knowledge sharing among subsurface teams, as the digital twin provides a common reference model that all can draw insights from.
- **Generalization:** While applied to a waterflood in a clastic reservoir, the principles of the REROSIM method are broadly applicable. Any reservoir where well interactions play a role (waterfloods, EOR floods, even some unconventional pad interactions) could benefit from a connectivity-based modeling approach. The success in this case study suggests that other brownfields could deploy similar digital twin frameworks to quickly realize production gains with relatively low implementation cost (since it leverages existing data and avoids intensive geological remodeling). It essentially turns abundant historical data into an asset for future optimization.

In conclusion, the REROSIM digital twin marks a step change in how we approach reservoir surveillance and optimization. It embodies the concept of a "smart reservoir" – one that is continuously understood and efficiently controlled through a synergy of physics, data, and automation. The Changqing Oilfield case



demonstrated tangible improvements in performance and has laid the groundwork for full-field rollout and replication in other assets.

This work contributes to the industry's ongoing digital transformation by providing a clear example of marrying petroleum engineering domain knowledge with advanced data analytics to solve real-field problems. It validates that with the right modeling strategy, we can greatly amplify the value of field data, moving from reactive management to autonomously optimized operations.

**Future Work:** Building on this successful pilot, future efforts will focus on scaling the digital twin to the entire Changqing Oilfield, incorporating more complex processes (such as polymer rheology and possibly CO<sub>2</sub>-WAG which is being tested in some blocks), and integrating surface network models for a complete production system optimization. Additionally, the use of emerging technologies like real-time distributed fiber optic sensing (for injection/production profiling) could feed into the twin, further enhancing its fidelity. On the algorithm side, multi-objective optimization (considering, for example, maximizing oil while minimizing carbon emissions of operations) could be explored, reflecting the evolving priorities in field development.

Ultimately, the journey of this project underscores a key lesson: data-driven digital solutions, grounded in physical reality, can unlock significant incremental value in reservoir management, even in assets that have been producing for decades. The tools and approaches presented here will be part of the next generation of reservoir management practices, enabling operators to meet energy demands more efficiently, economically, and sustainably.

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